IQRASEC ACADEMY

HTTP in Detail Report

WAPT Phase 1

Task 1 Web Fundamentals

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Report date

3 October 2026

Learning platform

TryHackMe

Description:

This report explains HTTP communication and the request exercises in HTTP in Detail room. It covers URLs, methods, status codes, headers and cookies, with spaces for practical results and screenshots.

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1. Objective

To interpret HTTP requests and responses, distinguish HTTP from HTTPS, identify URL components and parameters, and understand how methods, headers, status codes and cookies affect web application behavior.

2. Concept and Understanding

2.1 HTTP and HTTPS

HTTP is the Hypertext Transfer Protocol used to exchange web resources between a client and a server. HTTPS protects HTTP communication using TLS, providing encryption, integrity and server authentication. It protects data in transit; it does not remove application vulnerabilities.

2.2 URL Components

https://example.com:443/blog?id=1#comments

The scheme is https, the host is example.com, the port is 443, the path is /blog and the query is id=1. The fragment comments identifies a position within the page and is not included in the HTTP request target. Credentials should not be placed in URLs because they can appear in logs or browser history.

2.3 Requests and Responses

A client sends a request describing the resource and intended action. The server processes it and returns a response. In HTTP/1.1, a request starts with the method, request target and version. A response starts with the version, status code and reason phrase. Headers follow the start line, and a blank line separates headers from an optional body.

GET /blog?id=1 HTTP/1.1
Host: example.com
Accept: text/html

This illustrative request retrieves blog content using the id query parameter. Host identifies the destination host, and Accept expresses a preferred response format. It is an explanatory example rather than captured lab traffic.

2.4 Parameters and Request Bodies

Query parameters appear after the question mark in the request target. Body parameters carry submitted data separately from the URL. Form fields can use application/x-www-form-urlencoded, while APIs may use JSON. POST does not encrypt a body; HTTPS is needed to protect transport.

2.5 HTTP Methods

Method	Purpose
GET	Retrieve a representation of a resource.
POST	Submit data for processing, including form submissions.
PUT	Create or replace a representation at the specified target.
DELETE	Request removal of the target resource.

The application determines the behavior of each endpoint. Authorization must be checked for sensitive read, update and deletion operations regardless of the method used.

2.6 HTTP Status Codes

Code	Meaning
200 OK	The request succeeded.
201 Created	A resource was created.
301 Moved Permanently	The resource has a permanent new location.
302 Found	The client is directed to a temporary alternative location.
400 Bad Request	The server cannot process the request because of a client error.
401 Unauthorized	Valid authentication credentials are required.
403 Forbidden	The server refuses access.
404 Not Found	The requested resource was not found.
405 Method Not Allowed	The method is not supported for this target.
500 Internal Server Error	An unexpected server condition prevented processing.
503 Service Unavailable	The service is temporarily unable to handle the request.

2.7 Request and Response Headers

Host identifies the target host in HTTP/1.1. User-Agent identifies client software, Accept-Encoding advertises supported compression and Cookie carries applicable stored cookies. Content-Length expresses the body size in bytes when used. Content-Type identifies the media type, Content-Encoding identifies an applied encoding, and Cache-Control specifies caching rules.

2.8 Cookies and Session State

HTTP is stateless: each request is processed without an inherent memory of earlier requests. Cookies can support session continuity and preferences. A response uses Set-Cookie to request cookie storage; subsequent applicable requests use Cookie. A session identifier may refer to state held on the server.

Cookie transmission is limited by attributes such as domain, path, expiry and Secure. HttpOnly restricts JavaScript access. Session tokens should be concealed in report screenshots.

3. Practical Work

Task 7 contains a demo HTTP request emulator with five exercises. Use the gear control for URI or body parameters where required. For each exercise, record the returned status and visible result instead of assuming a successful outcome.

3.1 GET Request to the Room Page

GET /room

Select GET and request /room. Examine the returned content and status.

3.2 GET Request with a Query Parameter

GET /blog?id=1

Select GET, enter /blog and add the URI parameter id with value 1. The query selects the requested blog item.

3.3 DELETE Request

DELETE /user/1

Select DELETE and target /user/1. Inspect the response to determine how the removal request was handled.

3.4 PUT Request with a Body Parameter

PUT /user/2
Body parameter: username=admin

Select PUT, target /user/2 and set the body parameter username to admin. Review the returned update result.

3.5 POST Login Request

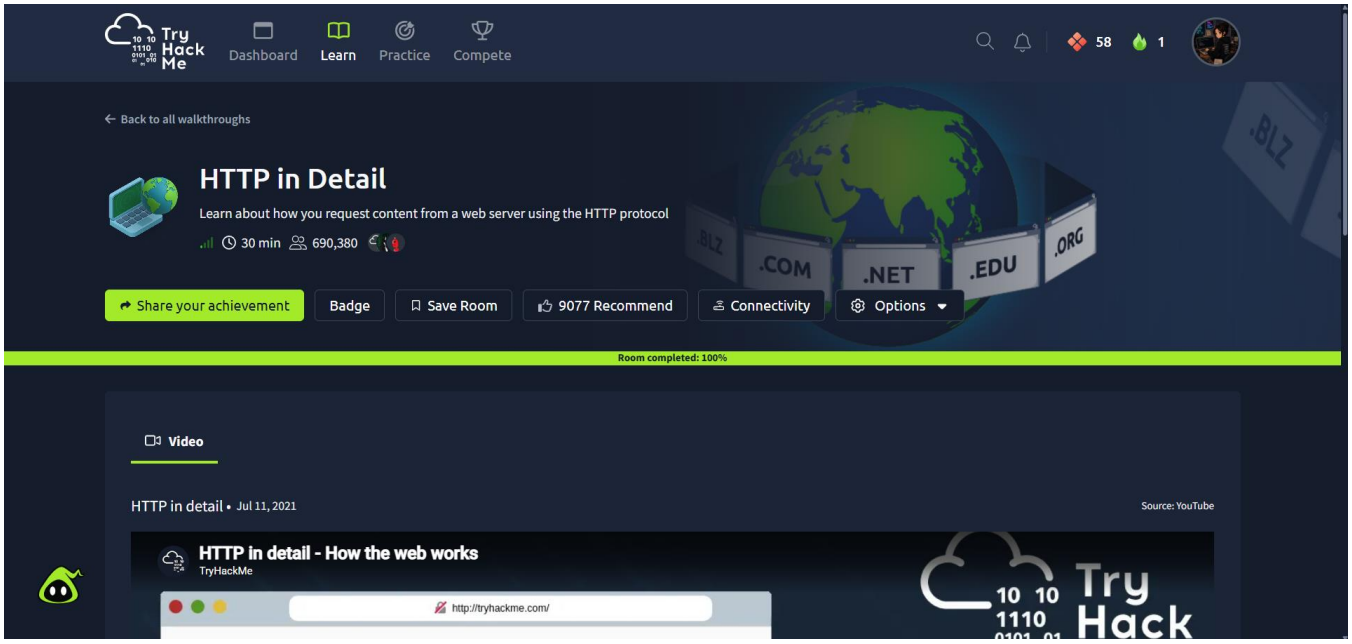
POST /login
Body parameters: username=thm and password=letmein

Submit the training credentials supplied by the room as body parameters. These are demo values, not personal account credentials. Review the login response.

4. Tools Used

TryHackMe provides the learning material. A web browser accesses the room, and its built-in request emulator supports the five practical exercises. Browser Developer Tools can inspect requests, responses and cookies when completing the cookie exercise; include that tool in your evidence only if you used it.

5. Evidence



6. Result

The exercises cover retrieval, query selection, deletion, updating and login submission. Actual outcomes and completion status should be recorded in Section 3 and supported by Figures 1 to 6 before submission.

7. Key Learnings

Methods describe intended operations, while parameters identify or supply data. Response codes, headers and bodies must be examined together. Cookies support session continuity, and HTTPS protects communication in transit. Understanding these fields prepares a tester to inspect and modify traffic using Burp Suite.

8. References

TryHackMe. HTTP in Detail. https://tryhackme.com/room/httpindetail

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